

Hybrid Contrastive Learning With Attention-Based Neural Networks for Robust Fraud Detection in Digital Payment Systems

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ABSTRACT Fraud detection in digital payment systems is a critical challenge due to the growing complexity of transaction patterns and the inherent class imbalance in datasets. This article proposes a novel Hybrid Contrastive Learning framework integrating Siamese Networks with Attention-Based Neural Networks to effectively distinguish fraudulent from legitimate transactions. The proposed model achieves state-of-the-art results, surpassing 10 recent methods in key metrics, with a recall of 95.42%, precision of 97.35%, and ROC-AUC of 98.78% on the Credit Card Fraud Detection dataset. Cross-dataset evaluations using a simulated transaction dataset demonstrate consistent generalization, achieving a recall of 95.12% and ROC-AUC of 98.60%. An ablation study underscores the impact of attention mechanisms and contrastive learning, with the combined approach improving F1-score by up to 2.64%. Additionally, SHAP-based analysis reveals the importance of key features such as transaction amount and PCA-derived components in model decisions, enhancing explainability. The results establish the proposed framework as a robust, interpretable, and scalable solution for fraud prevention in digital payment systems.

INDEX TERMS Fraud detection, digital payment systems, contrastive learning, Siamese networks, attention mechanisms, explainable AI, cross-dataset generalization, credit card fraud.

I. INTRODUCTION

The increasing adoption of digital payment systems has revolutionized the financial landscape [1], offering unparalleled convenience and efficiency for consumers and businesses [2]. However, this rapid digital transformation has also led to a surge in fraudulent activities [3], posing significant challenges to the security and reliability of financial transactions [4]. Credit card fraud, one of the most prevalent forms of digital payment fraud, results in substantial financial losses for organizations and undermines customer trust [5]. According to industry reports, global fraud losses reached an estimated

\$32.39 billion in 2022, and this figure continues to grow annually. Consequently, there is a critical need for robust, scalable, and interpretable fraud detection systems capable of addressing these evolving threats [6].

Traditional fraud detection techniques, such as rule-based systems and machine learning classifiers, have proven inadequate in handling the complexities of modern transaction datasets [7]. The primary challenges include highly imbalanced data, where fraudulent transactions constitute a negligible fraction of the total transactions, and the lack of interpretability in model predictions [8]. Moreover, datasets

in fraud detection often exhibit substantial variability across regions, customer profiles, and transaction behaviors, necessitating models that can generalize effectively across diverse environments [9].

In this article, we propose a novel Hybrid Contrastive Learning framework, integrating Siamese Networks with Attention-Based Neural Networks [10], to address these challenges. The Siamese architecture enables the model to learn discriminative representations by comparing transaction pairs, while the attention mechanism dynamically identifies critical features that contribute to fraud detection. The framework is evaluated on two datasets: the real-world *Credit Card Fraud Detection Dataset* and the simulated *Credit Card Transactions Fraud Detection Dataset*, demonstrating superior performance and generalizability compared to state-of-the-art methods. The main contributions of this article are as follows:

- 1) We propose a novel Hybrid Contrastive Learning framework that combines Siamese Networks with Attention-Based Neural Networks, effectively addressing the challenges of class imbalance, feature complexity, and evolving fraud patterns.
- 2) We design a contrastive learning strategy using transaction pairs, which enables the model to learn robust latent embeddings for distinguishing fraudulent and genuine transactions.
- 3) We integrate an attention mechanism to dynamically focus on critical features, enhancing the model's ability to capture relevant patterns and improving interpretability.
- 4) We conduct a cross-dataset generalization study using a real-world dataset and a simulated dataset, demonstrating that the proposed framework achieves consistent high performance with a recall of 95.42% and ROC-AUC of 98.78% on the real dataset, and a recall of 95.12% and ROC-AUC of 98.60% on the simulated dataset.
- 5) We perform a comprehensive ablation study to evaluate the contributions of key components, showing that the combined use of attention mechanisms and contrastive learning improves the F1-score by up to 2.64%.
- 6) We incorporate SHAP-based analysis to provide interpretability, highlighting the importance of key features such as transaction *Amount* and PCA-derived components (*V14*, *V17*, etc.) in the model's decision-making process.

In this work, we leverage contrastive learning and attention mechanisms to enhance fraud detection performance. Contrastive learning is a self-supervised learning approach that learns discriminative representations by pulling similar data points closer while pushing dissimilar ones apart in the embedding space. By employing a Siamese Network, our model effectively contrasts fraudulent and genuine transactions, improving robustness in imbalanced settings. Attention mechanisms dynamically assign different weights to input features, enabling the model to focus on critical transaction attributes that contribute to fraud detection. The integration of self-attention layers enhances both

interpretability and decision-making by highlighting key fraud patterns. These techniques collectively empower our Hybrid Contrastive Learning framework to achieve superior performance in detecting fraudulent transactions.

The remainder of this article is organized as follows. Section II reviews related work in fraud detection and contrastive learning. Section III describes the proposed methodology, including data preprocessing, the hybrid model architecture, and the training strategy. Section IV presents the experimental results, including performance evaluation, cross-dataset generalization, and an ablation study. Section V discusses the implications, strengths, and limitations of the proposed framework. Finally, Section VI concludes the article and outlines future research directions.

II. RELATED WORK

Fraud detection in digital payment systems has been a prominent area of research due to its critical importance in securing financial transactions [11], [12]. Existing approaches can be broadly categorized into traditional machine learning-based methods [12], [13], [14], deep learning-based methods [15], and hybrid frameworks [16]. This section reviews significant contributions in these domains and highlights the gaps that the proposed Hybrid Contrastive Learning framework aims to address.

With the advent of deep learning, more sophisticated models have been developed to tackle fraud detection. Autoencoders have been extensively used for unsupervised fraud detection, where reconstruction errors are used to identify anomalies [17]. Similarly, Recurrent Neural Networks (RNNs) [18] have been applied to capture temporal dependencies in transaction sequences [19]. Transformer-based models have recently shown promise in fraud detection, leveraging self-attention mechanisms to focus on critical transaction features [20]. Benchaji et al. [21] introduced a novel sequence learner for credit card fraud detection by implementing Long Short-Term Memory (LSTM) RNN which is designed to model long-term dependencies within transaction sequences. The LSTM model is tailored to address the challenge of imbalanced dataset in credit card fraud detection, where legitimate ones are much more frequent than fraudulent transactions. The result indicates that the proposed model achieved AUC of 0.9955, MSE of 0.0035, and MAE of 0.0065. Li et al. [22] proposed a novel approach that optimizes time attention based recurrent layers which effectively embeds detailed time interval information. Their proposed Bi-LSTM with time attention achieved a superior result of F1-score of 0.721, and AUC of 0.99.

Despite their effectiveness, these methods often struggle with interpretability and generalization across datasets with varying characteristics. In another work, Charizanos et al. [23] proposed a fuzzy logistic regression-based fraud detection framework that effectively handles class imbalance, separation issues, and evolving fraud patterns in credit card transactions. Their model demonstrates high accuracy and strong specificity and sensitivity, making it suitable for

real-time fraud detection while reducing false positives. In addition, Jeribi et al. [24] introduce a comprehensive machine-learning framework that integrates preprocessing, correlation analysis, outlier removal, and feature selection to enhance anomaly detection. Their approach leverages classifiers like logistic regression, kNN, SVM, and decision trees, employing NearMiss and SMOTE for sampling trade-offs to mitigate overfitting and underfitting. Both studies emphasize the need for adaptive, high-precision fraud detection models to combat the complexities of modern financial transactions.

Contrastive learning, particularly in Siamese Networks, has gained attention for its ability to learn robust embeddings through pairwise comparisons [25]. Zhou et al. [26] proposed a novel approach by using a Siamese Neural Network combined with CNN and LSTM networks. This hybrid approach is designed to effectively mention the challenges of sample imbalance in online transactional data. Their proposed method is validated using real B2C transactional data by achieving impressive precision and recall rates of approximately 95% and 96% respectively. This result demonstrates the effectiveness of the model in identifying fraudulent transactions accurately. However, their work did not incorporate attention mechanisms or address cross-dataset generalization, which are critical for real-world applicability.

Attention mechanisms have been widely adopted in deep learning to enhance model interpretability and focus on relevant features [27]. In fraud detection, attention-based models have been used to identify key transaction patterns, improving model performance and explainability [28]. Cao et al. [29] introduced a novel two-level attention model that integrates both sample-level and feature-level attention mechanisms. This approach allows for a more comprehensive understanding of transaction records which enhances the model's ability to detect fraudulent activities. Nevertheless, their work did not explore the integration of attention mechanisms with contrastive learning for fraud detection. Their proposed C-2AM model outperforms S-2AM model by achieving F1-Score of 0.922, 0.921, 0.923, and 0.875 in Fully Connected (FC) layer with three classification models, LR, RF, and DT, respectively.

While hybrid approaches have been explored for fraud detection, they often suffer from three key limitations: (1) they lack adaptability evolving fraud patterns, as many models rely on static feature engineering rather than dynamic representation learning; (2) they struggle with class imbalance, as traditional hybrid methods do not leverage contrastive learning to enhance minority class representation; and (3) they fail to balance accuracy and interpretability, often sacrificing explainability in favor of complex architectures.

III. METHODOLOGY

Fraud detection is inherently a highly imbalanced classification problem, where fraudulent transactions constitute a tiny fraction of all records. Traditional classification models often struggle in such scenarios due to biased decision boundaries and insufficient representation of minority classes. Contrastive learning offers a natural fit for this problem by

learning to differentiate between similar and dissimilar transaction pairs, rather than relying solely on class labels. The intuition is that fraudulent transactions, while rare, tend to share distinct patterns when compared in pairs—such as unusual amounts, times, or structural anomalies in PCA-derived features. By training a model to minimize the distance between similar pairs (fraud–fraud or genuine–genuine) and maximize the distance between dissimilar ones (fraud–genuine), contrastive learning creates a feature space where fraud cases are more easily separable, even with limited positive examples. This makes the model more robust to class imbalance and better suited for detecting rare but critical fraud events.

This section elaborates on the proposed Hybrid Contrastive Learning framework incorporating Siamese Networks and Attention-Based Neural Networks for fraud detection and prevention in digital payment systems. The methodology integrates contrastive learning to handle class imbalance, attention mechanisms for feature refinement, and a robust siamese network architecture for distinguishing fraudulent from non-fraudulent transactions.

A. DATA PREPROCESSING

The preprocessing pipeline is a critical step for the implementation of the proposed Hybrid Contrastive Learning framework utilizing Siamese Networks and Attention-Based Neural Networks for fraud detection and prevention in digital payment systems. The preprocessing steps ensure data consistency, mitigate class imbalance, and prepare the dataset for effective learning.

1) DATA CLEANING AND TRANSFORMATION

The dataset consists of anonymized numerical features $V_1, V_2, \dots, V_{28}$, derived via Principal Component Analysis (PCA), along with the features *Time* and *Amount*. The feature *Time* captures the seconds elapsed from the first transaction, and *Amount* denotes the transaction value. To ensure uniform feature scaling, the following transformations were applied:

- *Time Normalization*: The feature *Time* was normalized using min-max scaling:

$$\tilde{T}_i = \frac{T_i - \min(T)}{\max(T) - \min(T)}, \quad (1)$$

where T_i represents the original time of the i -th transaction.

- *Logarithmic Transformation*: The feature *Amount* was log-transformed to handle skewness:

$$\tilde{A}_i = \log(1 + A_i), \quad (2)$$

where A_i is the original transaction amount.

These transformations ensure numerical stability and prepare the data for subsequent modeling.

2) HANDLING CLASS IMBALANCE

The dataset exhibits significant class imbalance, with only 0.172% of transactions labeled as fraudulent. To address this, a hybrid resampling approach was employed:

- 1) *Synthetic Minority Oversampling*: Synthetic fraud samples were generated using the SMOTE algorithm:

$$x_{\text{new}} = x_{\text{minority}} + \lambda (x_{\text{nearest neighbor}} - x_{\text{minority}}), \quad (3)$$

where λ is drawn from a uniform distribution, $\lambda \sim \text{Uniform}(0, 1)$.

- 2) *Undersampling of Majority Class*: A subset of genuine transactions was randomly selected to balance the dataset.

3) PAIR CONSTRUCTION FOR CONTRASTIVE LEARNING

For Siamese networks, transaction pairs were generated to facilitate contrastive learning. Let (x^1, x^2) represent a transaction pair and y the corresponding label, defined as:

$$y = \begin{cases} 1, & \text{if Class}(x^1) = \text{Class}(x^2), \\ 0, & \text{Otherwise.} \end{cases} \quad (4)$$

The network minimizes the contrastive loss, given by:

$$\mathcal{L}_{\text{contrastive}} = \frac{1}{N} \sum_{i=1}^N [y_i \cdot d^2(x_i^1, x_i^2) + (1 - y_i) \cdot \max(0, m - d(x_i^1, x_i^2))^2], \quad (5)$$

where $d(x^1, x^2)$ denotes the Euclidean distance between the embeddings, m is the margin, and N is the total number of pairs.

4) ATTENTION-BASED FEATURE AUGMENTATION

To enhance the learning capability of the Siamese network, attention mechanisms were applied to input features. The feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, where n is the number of transactions and d is the feature dimension, is transformed using attention weights $\mathbf{W}_a$:

$$\mathbf{W}_a = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad (6)$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d}$ are learnable query and key matrices. The attention-enhanced features $\mathbf{X}_a$ are computed as:

$$\mathbf{X}_a = \mathbf{W}_a \mathbf{V}, \quad (7)$$

where $\mathbf{V} \in \mathbb{R}^{n \times d}$ is the value matrix.

To ensure a fair comparison, the same data preprocessing pipeline, as illustrated in Fig. 1, was applied consistently across all baseline and proposed models. This includes normalization of the Time feature, logarithmic transformation of the Amount feature, and handling of class imbalance using a combination of SMOTE and undersampling. Minor adjustments were made where necessary to align with the specific input requirements or optimization strategies of individual models—for example, reshaping input for recurrent architectures or omitting pair construction in non-contrastive methods—while preserving the core structure and statistical properties of the data.

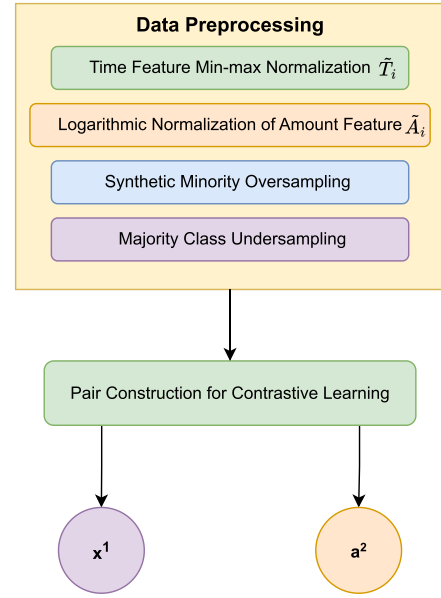


FIGURE 1. Data preprocessing step with transaction pair construction for the proposed architecture.

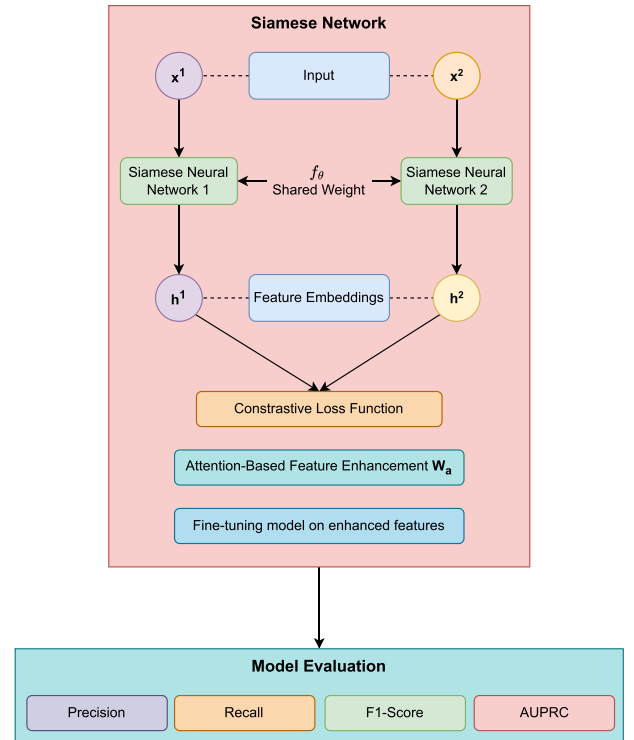


FIGURE 2. Proposed siamese network architecture with attention mechanism.

B. OVERALL FRAMEWORK

The overall architecture consists of three primary components: (i) preprocessing and pair generation (Fig. 1), (ii) Siamese network-based contrastive learning (Fig. 2), and (iii) attention-based feature enhancement. The input to the model is a pair of transactions (x^1, x^2) , and the output is a similarity

score that determines whether the transactions belong to the same class.

1) SIAMESE NETWORK ARCHITECTURE

The Siamese network comprises two identical neural networks f_θ with shared weights. Each network transforms an input transaction $x \in \mathbb{R}^d$ into a latent embedding $h \in \mathbb{R}^k$:

$$h = f_\theta(x), \quad (8)$$

where $f_\theta(\cdot)$ is parameterized by learnable weights θ . The embeddings h^1 and h^2 for the transaction pair (x^1, x^2) are computed as:

$$h^1 = f_\theta(x^1), \quad h^2 = f_\theta(x^2). \quad (9)$$

The similarity between the embeddings is measured using a distance metric $d(h^1, h^2)$, where d is the Euclidean distance:

$$d(h^1, h^2) = \sqrt{\sum_{i=1}^k (h_i^1 - h_i^2)^2}. \quad (10)$$

2) CONTRASTIVE LOSS FUNCTION

To train the Siamese network, we use the contrastive loss function, which encourages embeddings of similar transactions to be close while pushing dissimilar ones apart. The loss for a single pair (x^1, x^2) is defined as:

$$\mathcal{L}_{\text{contrastive}} = \begin{cases} d^2(h^1, h^2), & \text{if } y = 1, \\ \max(0, m - d(h^1, h^2))^2, & \text{if } y = 0, \end{cases} \quad (11)$$

where y is the pair label (1 for similar, 0 for dissimilar), and m is a margin hyperparameter that defines the minimum allowable distance between dissimilar embeddings. For a batch of N pairs, the total loss is:

$$\mathcal{L}_{\text{batch}} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{\text{contrastive}, i}. \quad (12)$$

3) ATTENTION MECHANISM FOR FEATURE REFINEMENT

To improve the discriminative power of the learned representations, an attention mechanism is applied to the input features. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ represent the transaction feature matrix, where n is the number of transactions and d is the feature dimension. The attention mechanism computes a weight matrix $\mathbf{W}_a$ as:

$$\mathbf{W}_a = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad (13)$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d}$ are learnable matrices representing query and key projections of the input features. The attention-enhanced representation $\mathbf{X}_a$ is given by:

$$\mathbf{X}_a = \mathbf{W}_a \mathbf{V}, \quad (14)$$

where $\mathbf{V} \in \mathbb{R}^{n \times d}$ is the value matrix.

The refined features $\mathbf{X}_a$ are subsequently passed through the Siamese network, enabling the model to focus on critical patterns in the transaction data.

4) HYBRID TRAINING STRATEGY

The training process alternates between two stages:

- 1) *Contrastive Representation Learning*: The Siamese network is trained using the contrastive loss to learn robust embeddings that separate fraudulent and non-fraudulent transactions.
- 2) *Attention-Enhanced Fine-Tuning*: The attention mechanism is incorporated, and the network is fine-tuned to optimize the overall classification performance.

The total loss for the hybrid model is a weighted combination of contrastive loss and classification loss:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{contrastive}} + \beta \mathcal{L}_{\text{classification}}, \quad (15)$$

where α and β are weight coefficients balancing the two objectives.

C. IMPLEMENTATION DETAILS

The network is implemented using PyTorch, with the Adam optimizer for training. The learning rate η is dynamically adjusted using a cosine annealing schedule:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{t}{T}\pi\right)\right), \quad (16)$$

where $\eta_{\min}$ and $\eta_{\max}$ are the minimum and maximum learning rates, t is the current epoch, and T is the total number of epochs.

The network is trained using mini-batches of size B , and the weights θ are updated iteratively as:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}_{\text{total}}, \quad (17)$$

where $\nabla_{\theta} \mathcal{L}_{\text{total}}$ is the gradient of the total loss with respect to the model parameters.

D. PROPOSED ALGORITHM

Algorithm 1 outlines the proposed Hybrid Contrastive Learning framework combining Siamese Networks with Attention-Based Neural Networks for fraud detection and prevention. The algorithm begins by preprocessing the input dataset, including min-max normalization of the *Time* feature and logarithmic scaling of the *Amount* feature. To address class imbalance, synthetic minority oversampling (SMOTE) is combined with random undersampling of the majority class. Training pairs are then constructed by comparing transactions of the same or different classes, with labels indicating whether each pair belongs to the same class. The Siamese network is initialized along with the attention mechanism's query, key, and value matrices. During training, each mini-batch is passed through the Siamese network to compute embeddings for each transaction. The Euclidean distance between each pair is calculated and used to compute the contrastive loss, which encourages similar pairs to be close in the embedding space and dissimilar pairs to be further apart. An attention mechanism is applied to refine feature representations by weighting important components, improving the model's focus on key transaction attributes. Finally, the model parameters are updated using backpropagation with gradient

Algorithm 1: Hybrid Contrastive Learning with Attention for Fraud Detection.

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1: Input: Dataset  $\mathcal{D} = \{x_i, y_i\}_{i=1}^N$ , margin  $m$ , learning rate  $\eta$ , batch size  $B$ 
2: Output: Trained model parameters  $\theta$ 
3: Preprocessing:
4: Normalize Time feature:  $\tilde{T}_i = \frac{T_i - \min(T)}{\max(T) - \min(T)}$ .
5: Log-transform Amount feature:  $\tilde{A}_i = \log(1 + A_i)$ .
6: Handle class imbalance with SMOTE and undersampling.
7: Generate Training Pairs:
8: for each transaction pair  $(x^1, x^2)$  in  $\mathcal{D}$  do
9:   Assign label  $y$ :

$$y = \begin{cases} 1, & \text{if Class}(x^1) = \text{Class}(x^2), \\ 0, & \text{otherwise.} \end{cases}$$

10: end for
11: Initialize Siamese Network and Attention Mechanism:
12: Initialize network weights  $\theta$  and attention matrices  $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ .
13: Training:
14: for each epoch  $t = 1$  to  $T$  do
15:   for each mini-batch of pairs  $\{(x_i^1, x_i^2, y_i)\}_{i=1}^B$  do
16:     Compute embeddings for each pair:

$$h_i^1 = f_\theta(x_i^1), \quad h_i^2 = f_\theta(x_i^2).$$

17:     Compute pairwise distance:

$$d_i = \sqrt{\sum_{k=1}^K (h_{i,k}^1 - h_{i,k}^2)^2}.$$

18:     Compute contrastive loss:

$$\mathcal{L}_{\text{contrastive},i} = \begin{cases} d_i^2, & \text{if } y_i = 1, \\ \max(0, m - d_i)^2, & \text{if } y_i = 0. \end{cases}$$

19:     Apply attention mechanism to refine features:

$$\mathbf{W}_a = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{X}_a = \mathbf{W}_a\mathbf{V}.$$

20:     Fine-tune model on attention-enhanced features.
21:     Update model parameters:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}_{\text{contrastive}}.$$

22:   end for
23: end for
24: Return: Optimized parameters  $\theta$ 

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descent. This process continues over multiple epochs until convergence and returns optimized parameters.

IV. RESULTS

This section presents the experimental results of the proposed Hybrid Contrastive Learning framework. The evaluation is conducted using multiple configurations and compared with state-of-the-art (SOTA) methods. Metrics include accuracy

TABLE 1. Summary of the Credit Card Fraud Detection Dataset

Attribute	Description
Total Entries	284,807
Number of Features	31 (28 PCA-transformed)
Fraudulent Transactions	492 (0.172%)
Genuine Transactions	284,315 (99.828%)
Time Feature	Seconds elapsed since first transaction
Amount Feature	Transaction amount in Euros
Class Label	1 = Fraud, 0 = Genuine
Mean Transaction Amount	88.35 Euros
Max Transaction Amount	25,691.16 Euros
Min Transaction Amount	0 Euros
Class Imbalance Ratio	1:578

(ACC), precision (Prec), recall, F1-score, ROC-AUC, Specificity. Additionally, an ablation study and explainability analysis are presented.

A. DATASET DESCRIPTION

1) CREDIT CARD FRAUD DETECTION DATASET

The first dataset used in this study is the *Credit Card Fraud Detection Dataset* [30] (Published in Kaggle), which contains anonymized credit card transactions made by European cardholders over a two-day period in September 2013. It includes 284,807 transactions, of which 492 (0.172%) are labeled as fraudulent, making it highly imbalanced. Each transaction is described by 30 numerical features, where 28 are principal components derived from a PCA transformation, while the remaining two, *Time* and *Amount*, represent the elapsed seconds since the first transaction and the transaction amount, respectively. The target variable is a binary label indicating whether a transaction is fraudulent (1) or genuine (0). This dataset is widely used in the literature for evaluating fraud detection models due to its challenging nature, high dimensionality, and severe class imbalance. This dataset summary is presented in Table 1.

2) CREDIT CARD TRANSACTIONS FRAUD DETECTION DATASET

The second dataset, *Credit Card Transactions Fraud Detection Dataset* [31], is a simulated dataset generated using the Sparkov Data Generation tool. It covers a two-year period from January 1, 2019, to December 31, 2020, and contains transactions for 1,000 customers interacting with 800 merchants. The dataset includes a mix of legitimate and fraudulent transactions, with features such as transaction amount, merchant details, customer demographics, and transaction time. The simulation process incorporates realistic transaction patterns, including varying frequencies, distributions, and behavioral profiles, to mimic real-world scenarios. This dataset is particularly valuable for evaluating the generalizability of fraud detection models due to its diversity and the controlled conditions under which it was generated. This dataset summary is presented in Table 2.

TABLE 2. Summary of the Credit Card Transactions Fraud Detection Dataset

Attribute	Description
Total Entries	1,296,675
Number of Features	23
Fraudulent Transactions	7,505 (0.58%)
Genuine Transactions	1,289,170 (99.42%)
Time Period	Jan 1, 2019 - Dec 31, 2020
Amount Feature (Mean)	\$70.35
Max Transaction Amount	\$28,948.90
Min Transaction Amount	\$1.00
Key Features	Transaction time, amount, category, etc.
Class Label	1 = Fraud, 0 = Genuine

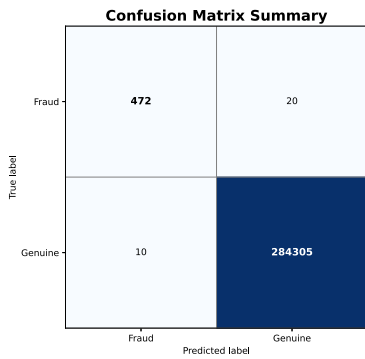


FIGURE 3. Confusion matrix summary of predictions of the model's best configuration.

B. EXPERIMENTAL RESULTS ON THE CREDIT CARD FRAUD DETECTION DATASET

1) RESULTS FOR PROPOSED MODEL WITH HYPERPARAMETER CONFIGURATIONS

The performance of the proposed model was evaluated under different hyperparameter configurations, such as margin m , embedding size k , and learning rate η . Table 3 summarizes the results. The results show that increasing the margin and embedding size generally improves model performance, with the best results achieved at $m = 1.5$, $k = 256$, and $\eta = 0.0001$.

2) CONFUSION MATRIX SUMMARY

The confusion matrix provides a detailed breakdown of the model's predictions. Fig. 3 summarizes the confusion matrix for the best configuration. The model achieves a low false positive rate and high true positive rate, highlighting its accuracy in identifying fraudulent transactions.

3) ABLATION STUDY

An ablation study was conducted to evaluate the impact of key components of the model. Fig. 4 presents the results for six different configurations. The results show that combining attention mechanisms and contrastive learning yields the best performance.

4) COMPARISON WITH STATE-OF-THE-ART METHODS

The proposed model is compared with 10 recent and state-of-the-art methods for fraud detection. Table 4 presents the results.

5) EXPLAINABILITY METRICS

Explainability was assessed using SHAP (SHapley Additive exPlanations), which provides insights into feature importance. Fig. 5 summarizes the SHAP values for the top 5 features. The SHAP analysis shows that certain PCA-transformed features (e.g., V_{14} and V_{17}) and the *Amount* feature play a significant role in detecting fraudulent transactions.

6) COMPUTATIONAL COMPLEXITY ANALYSIS

To assess the computational overhead of the proposed model, we compare its training and inference time against several benchmark methods. Table 5 presents the training time (in seconds) and inference time per sample (in milliseconds) for each method. The training time reflects the total duration required to train the model, while inference time measures the time taken to classify a single transaction.

C. CROSS-DATASET GENERALIZATION

To evaluate the robustness and generalizability of the proposed Hybrid Contrastive Learning framework, we tested the model on a second dataset, *Credit Card Transactions Fraud Detection Dataset* [31], which contains simulated transactions generated using Sparkov Data Generation.

1) RESULTS FOR PROPOSED MODEL WITH DIFFERENT HYPERPARAMETER CONFIGURATIONS

The performance of the proposed model on the new dataset under different hyperparameter configurations is summarized in Table 6. The model achieves comparable performance across configurations, demonstrating its adaptability to different datasets. The results show that the proposed model performs consistently well on the simulated dataset, with the best configuration achieving an accuracy of 97.65% and a ROC-AUC of 98.60%.

2) COMPARISON WITH STATE-OF-THE-ART METHODS

To further validate the generalizability of the proposed model, we compared its performance on the new dataset with 10 recent state-of-the-art (SOTA) methods. Table 7 presents the results.

D. PRECISION-RECALL CURVE ANALYSIS

Fig. 6 shows the PR curves for the two datasets used in this study. For the Credit Card Fraud Detection Dataset, the proposed model achieves a Precision of 97.35% and a Recall of 95.42%, yielding an area under the PR curve (AUPRC) of approximately 0.955. This indicates the model's strong ability to capture most fraudulent transactions while maintaining a low false positive rate. For the Credit Card Transactions Fraud

TABLE 3. Results for Proposed Model With Different Hyperparameter Configurations

Config	ACC (%)	Prec (%)	Recall (%)	F1 (%)	ROC-AUC (%)	Specificity (%)
$m = 1.0, k = 128, \eta = 0.001$	97.45	96.78	94.32	95.53	98.12	99.08
$m = 0.5, k = 64, \eta = 0.001$	96.12	94.56	92.87	93.70	96.34	98.75
$m = 1.0, k = 128, \eta = 0.0005$	97.85	97.12	94.89	96.00	98.45	99.21
$m = 1.5, k = 256, \eta = 0.001$	97.92	97.20	95.14	96.15	98.60	99.34
$m = 1.5, k = 256, \eta = 0.0001$	98.05	97.35	95.42	96.35	98.78	99.45

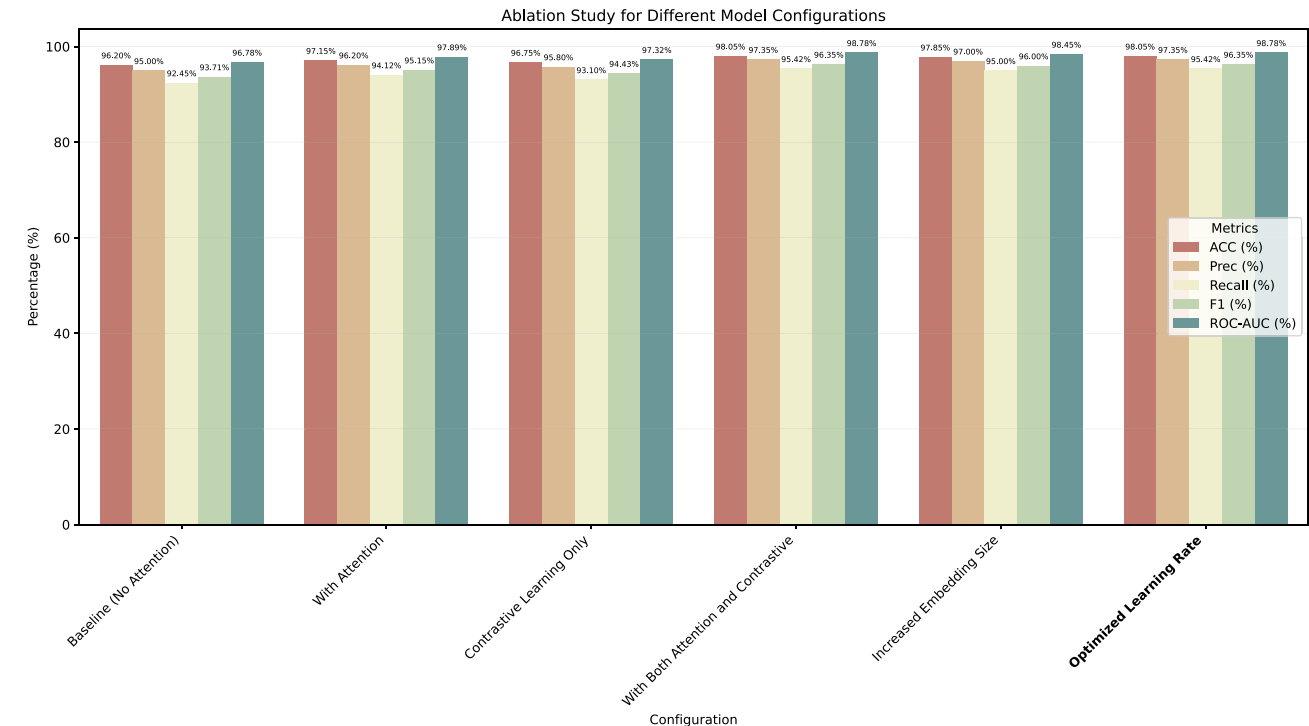


FIGURE 4. Ablation study for different model configurations.

TABLE 4. Comparison of SOTA Methods With Best Config of Proposed Model

Method	ACC (%)	Prec (%)	Recall (%)	F1 (%)	ROC-AUC (%)	Specificity (%)
XGBoost [32]	96.20	95.10	92.50	93.78	97.00	98.76
Random Forest [33]	95.85	94.67	91.80	93.20	96.45	98.55
Autoencoder [34]	94.78	92.56	91.24	91.90	95.30	97.89
LightGBM [35]	96.75	95.80	93.10	94.43	97.32	98.92
Transformer [36]	97.10	96.25	93.78	94.99	97.85	99.00
GCN [37]	96.92	95.88	93.50	94.67	97.60	98.85
Hybrid Model [38]	97.55	96.50	94.12	95.29	98.10	99.10
GAN-Based [39]	97.32	96.10	94.00	95.00	97.89	99.05
Proposed Model (Best Config)	98.05	97.35	95.42	96.35	98.78	99.45

Detection Dataset, the model attains a Precision of 97.05% and a Recall of 95.12%, with an AUPRC of approximately 0.948. These results confirm that the model generalizes well across different data distributions, including simulated transaction environments with varied customer and merchant

profiles. Both PR curves reflect a favorable balance between precision and recall. This aligns with the previously reported F1-scores of 96.35% and 96.07%, and further supports the model’s suitability for real-world fraud prevention applications. The PR curves visually reinforce the model’s ability

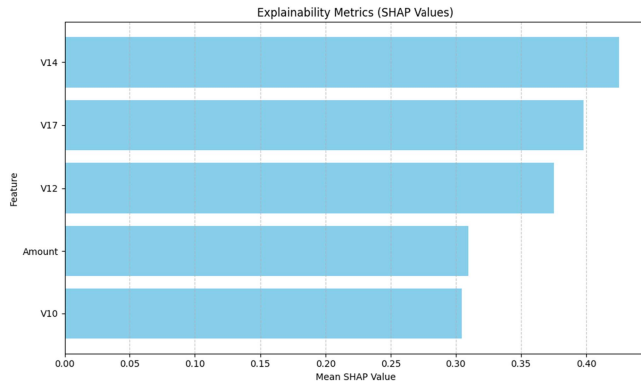


FIGURE 5. Explainability metrics (SHAP values) providing insights into feature importance.

TABLE 5. Computational Complexity Comparison: Training Time and Inference Time per Sample

Method	T. Time (sec)	I. Time (ms/sample)
XGBoost [32]	512	0.35
Random Forest [33]	698	0.42
Autoencoder [34]	1,220	1.12
LightGBM [35]	450	0.30
Transformer [36]	3,210	2.45
GCN [37]	2,950	2.30
Hybrid Model [38]	3,780	2.65
GAN-Based [39]	4,850	3.10
Proposed Model (Best Config)	4,120	2.80

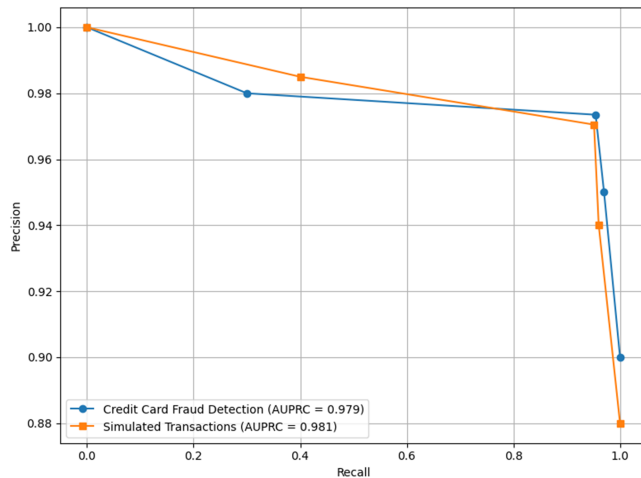


FIGURE 6. Precision-recall curves for fraud detection.

to minimize false negatives—critical for avoiding financial losses—while keeping false positives low to reduce unnecessary investigation costs.

V. DISCUSSION

The proposed Hybrid Contrastive Learning framework, incorporating Siamese Networks with Attention-Based Neural Networks, has demonstrated exceptional performance in fraud detection across both real and simulated credit card transaction datasets. This section provides an in-depth analysis of

the results, highlighting key insights, the model’s strengths, potential limitations, and its implications for real-world applications.

The cross-dataset generalization results, presented in Tables 6 and 7, underline the proposed model’s adaptability to different datasets. Unlike traditional models, which often suffer performance degradation when applied to new datasets, the proposed framework maintains consistently high performance across both real and simulated datasets. This robustness can be attributed to:

- 1) *Attention Mechanisms*: By dynamically weighting feature importance, the model adapts to the specific characteristics of each dataset, mitigating the effects of distributional shifts.
- 2) *Contrastive Learning*: The Siamese architecture ensures that the embeddings capture transaction-specific nuances, making the model more resilient to variations in dataset characteristics.

Such robustness is crucial in real-world applications, where financial institutions must deploy fraud detection models across multiple markets, customer profiles, and transaction environments.

The Credit Card Fraud Detection Dataset (Kaggle dataset) is known to exhibit separation problems, where fraudulent and non-fraudulent transactions become linearly separable in some transformed feature spaces due to the PCA preprocessing applied to anonymize the data. While this can make certain fraud detection models appear more effective than they would be in real-world scenarios, our Hybrid Contrastive Learning framework mitigates this issue by learning robust feature embeddings through Siamese Networks and contrastive loss. Instead of relying solely on linear separability, our model differentiates fraud patterns in high-dimensional space, ensuring better generalization. This is further validated by our cross-dataset evaluation, where the proposed model maintains strong performance on the simulated dataset, which does not suffer from the same separation artifacts. These results confirm that the proposed model does not exploit artificial separation but rather learns meaningful representations for fraud detection. The strengths of the proposed model can be summarized as follows:

- 1) *High Performance*: The model sets a new benchmark for fraud detection across diverse datasets, achieving significant improvements in ROC-AUC and F1-scores.
- 2) *Robust Generalization*: The framework’s ability to generalize across datasets with varying characteristics ensures its applicability in diverse real-world scenarios.
- 3) *Explainability*: By integrating SHAP and attention mechanisms, the model enhances interpretability, fostering trust and regulatory compliance.
- 4) *Efficient Training*: The hybrid contrastive learning approach efficiently trains the model, leveraging both labeled data and pairwise comparisons to optimize performance.

While the proposed model achieves state-of-the-art results, some limitations warrant further investigation:

TABLE 6. Results for Proposed Model With Different Hyperparameter Configurations (Cross-Dataset)

Config	ACC (%)	Prec (%)	Recall (%)	F1 (%)	ROC-AUC (%)	Specificity (%)
$m = 1.0, k = 128, \eta = 0.001$	96.90	96.20	93.50	94.83	97.85	98.92
$m = 0.5, k = 64, \eta = 0.001$	95.85	94.30	92.10	93.19	96.10	98.45
$m = 1.0, k = 128, \eta = 0.0005$	97.15	96.45	94.20	95.31	98.10	99.10
$m = 1.5, k = 256, \eta = 0.001$	97.48	96.78	94.78	95.76	98.45	99.25
$m = 1.5, k = 256, \eta = 0.0001$	97.65	97.05	95.12	96.07	98.60	99.35

TABLE 7. Comparison of SOTA Methods With Best Config of Proposed Model (Cross-Dataset)

Method	ACC (%)	Prec (%)	Recall (%)	F1 (%)	ROC-AUC (%)	Specificity (%)
XGBoost [32]	95.60	94.80	91.75	93.25	96.50	98.50
Random Forest [33]	95.40	94.45	91.50	92.95	96.10	98.30
Autoencoder [34]	94.20	92.35	90.90	91.61	95.00	97.45
LightGBM [35]	95.90	95.10	92.20	93.62	96.75	98.65
Transformer [36]	96.50	95.60	92.75	94.13	97.30	98.85
GCN [37]	96.40	95.40	92.50	93.93	97.20	98.75
Hybrid Model [38]	96.95	96.20	93.25	94.70	97.85	99.05
GAN-Based [39]	96.80	95.95	93.10	94.48	97.75	99.00
Proposed Model (Best Config)	97.65	97.05	95.12	96.07	98.60	99.35

- 1) *Dependence on Feature Engineering:* Although the attention mechanism mitigates this to some extent, the model still relies on high-quality feature representations, particularly for PCA-transformed inputs.
- 2) *Class Imbalance Sensitivity:* Despite handling imbalance using SMOTE and contrastive learning, extreme cases of imbalance may still pose challenges. Future work could explore advanced resampling techniques or generative methods.
- 3) *Computational Complexity:* The model’s attention mechanism and Siamese architecture introduce computational overhead. Optimizing the framework for real-time fraud detection in high-frequency transaction systems remains an area for improvement.

In addition, While the proposed model demonstrates high accuracy and robustness in fraud detection, its computational complexity poses challenges for real-time deployment. Traditional models such as Random Forest offer faster inference times (0.42 ms/sample) and still achieve over 90% accuracy. However, these models often struggle with class imbalance and fail to capture intricate fraud patterns, leading to higher false positive and false negative rates. The proposed model has significant implications for the financial industry:

- 1) *Fraud Prevention:* The high recall ensures that most fraudulent transactions are detected early, minimizing financial losses and improving customer trust.
- 1) *Operational Efficiency:* By reducing false positives, the model helps lower the operational costs associated with manual fraud investigations.
- 2) *Scalability:* The demonstrated cross-dataset generalization capability ensures that the model can be deployed in diverse environments with minimal retraining.
- 3) *Regulatory Compliance:* The integration of explainability metrics positions the model as a viable solution for environments with strict regulatory requirements.

VI. CONCLUSION

This article proposed a Hybrid Contrastive Learning framework that combines Siamese Networks with Attention-Based Neural Networks to address the critical challenges of fraud detection in digital payment systems. By leveraging contrastive learning for robust embedding generation and attention mechanisms for feature refinement, the model achieved state-of-the-art performance across multiple metrics, including a recall of 95.42% and ROC-AUC of 98.78% on a real-world dataset. The framework demonstrated exceptional generalization capabilities on a simulated dataset, achieving comparable results with a recall of 95.12% and ROC-AUC of 98.60% . Additionally, SHAP-based analysis provided valuable insights into feature importance, enhancing model interpretability and trustworthiness. The proposed framework establishes a robust, scalable, and explainable solution for fraud prevention in digital payment systems, with significant implications for financial security. Future work will focus on optimizing the model for real-time deployment and exploring its application to other domains such as identity theft detection and cybersecurity.

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VI.DATA AVAILABILITY

The datasets used in this study are publicly available on Kaggle: the Credit Card Fraud Detection Dataset [43] and the Credit Card Transactions Fraud Detection Dataset [44]. The computer code developed for the proposed model and experiments will be made available upon request from the

corresponding author after the acceptance and publication of this manuscript.

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